

Morse Decode (VK2ARH Version)

Based on 'Hamfist' Decoder by Jon Dawson

Construction Manual

Version 1.0



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Introduction

The 'Hamfist' Morse Decoder was chosen by the Manly Warringah Radio Society <https://www.mwrs.org.au/> as one of the 2026 build program projects. The key components of the selection were:

- Low cost using readily available components.
- Use of through hole devices wherever practical – to enable construction by novice/intermediate builders.
- Provide a practical piece of equipment that can be used in the shack or work bench to facilitate building skills, learning, experimentation and have fun both building and using the project.

This layout was designed to utilize through hole components and a self-contained 'PCB sandwich' construction to facilitate construction by beginners in Radio Club Buildathon style of events or by individuals who do not wish to use SMD technology. Housing in any form of 3D printed enclosure is at the builder's discretion.

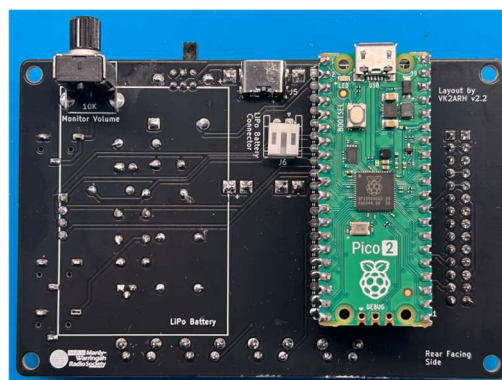


The design is based on the 'Hamfist' Morse Decoder developed by Jon Dawson and a great vote of thanks to Jon for his work and sharing his design for others to enjoy. This layout took Jon's design and laid it out onto a PCB, added an audio monitor circuit with volume control, a PTT (press to talk) circuit to key a companion morse code transmitter if required and a LiPo battery option. The design supports both the touch screen and push button user interfaces with the decoder software, and also supports any further development which chooses to use the SD card reader on the TFT display.

Full details of the design concept behind the decoder and the latest software can be found at Jon Dawson's Site: https://101-things.readthedocs.io/en/latest/ham_fist.html. This site provides an excellent series of videos outlining the theory of operation and design objectives. The software will no doubt continue to be updated with new and increased functionality as time progresses – so be sure to visit the site regularly for any updates.

Copies of this construction manual, PCB Gerber files and associated documentation for this project can be found at my GitHub site: <https://github.com/VK2ARH/>

The morse decoder is a relatively easy and quick construction project with a low component count that can be built in approx. on hour or less by an experienced builder. The project also lends itself to novice kit builders who under supervision or assistance from experienced builders will be able to complete the build.



The manual was written with a novice constructor in mind and is focused on a step-by-step instructional build process. Seasoned kit builders please ignore/excuse the detail in parts.

I recommend that you read the entire manual before you commence construction to gain an understanding of the process to be followed, the contents of this manual and familiarize yourself with the journey ahead. This manual outlines the building process using the MWRS kit which was put together as a 'Group Buy', but all components are readily available from electronics supply companies.

Richard VK2ARH / W5ARH

Kit Assembly

Equipment Needed:

The following equipment is recommended to build the kit:

- Soldering Iron / Station (preferably ESD protected with a fine tip, together with a method of venting fumes away from the soldering area).
- Solder: 0.8mm or smaller diameter 63% Sn / 37% Pb with 'no clean' flux is recommended, if you are comfortable soldering with Pb based solder.
- Solder flux can be an advantage in soldering some components but is not essential if you have a good quality solder with embedded flux.
- Wire cutters (fine precision type)
- Multimeter
- Phillips head screwdriver
- Modeling knife
- Solder sucker or solder wick if needed.
- A magnifying glass or higher power reading glasses can assist with checking the quality of your soldering.

Take the time to prepare your work area and layout your tools, keep it free of clutter. An organized work area helps you to focus on the build and not be distracted by searching for tools and components.



An example of a cleared and ordered work bench ahead of the kit build. Individual equipment will vary.
(My workbench isn't always that tidy 😊)

General Build Project Construction Tips

Whilst it makes no difference electronically which way the resistors (in this case) are mounted, good practice is to mount each device with the same orientation. By doing so it makes checking the value of components much easier.

For capacitors or devices with identification marks on them, I try to orient them all in the same direction unless their view is being obstructed by another component. When this happens, I orient them where possible, in a direction that enables them to be read easily.

When you place the through hole components on the PCB hold the component down against the PCB and spread the component leads so that when the board is turned over to solder, the components remain in position. Restrict the number of components you place at any one time to avoid a 'rats nest' of leads that will interfere with your soldering iron tip and the solder whilst soldering.

Once you have soldered the components – remove the leads as close as possible to the solder joint using a small pair of wire cutters. This process should be performed after installation of each component group.

I place my finger on the end of the lead when snipping the lead to absorb the 'snap' shock of the cut – reducing stress on the recently soldered join (**but not on static sensitive devices**). It also stops the lead 'flying' away when you cut it. If the lead is soft and you cut slowly this technique is probably not necessary, but it's something to consider.

Don't use excessive heat whilst soldering – this has the potential to damage components and destroy the PCB tracks. I like to tin the solder iron tip and then use the 3 second rule ensuring that the soldering iron is not in contact with the device for more than 3 seconds at a time when soldering each lead. If you need longer, allow the lead to cool and return to complete or improve the solder joint.

Flux or flux paste is not usually needed if you are using a good quality solder with incorporated flux, but it does help if you have trouble with solder flowing. Choose a 'no clean' flux if you have the opportunity as this makes for a cleaner result. If necessary, boards can be cleaned with an old toothbrush and isopropyl alcohol or for those looking for a 'Rolls Royce Solution' access to and cleaning with isopropyl alcohol in an ultrasonic cleaner produces great results.

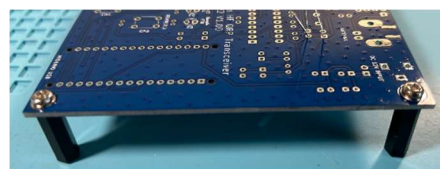
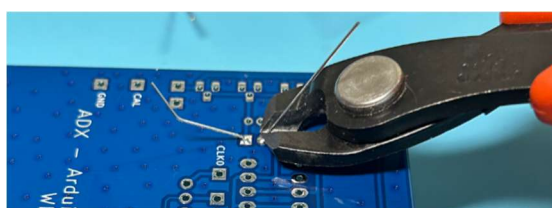
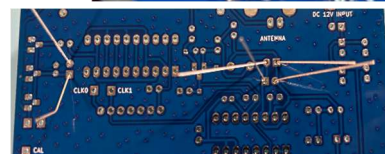
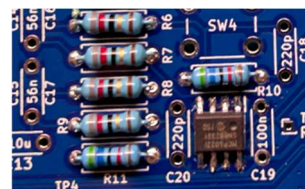
Hint: Isopropyl alcohol can be purchased from the pharmacy or hardware store. The higher the alcohol content the better. Scrubbing and rinsing with a mild dishwashing detergent and water and then drying with a hair dryer produces good results removing the sticky residue that sometimes is present after cleaning with the isopropyl alcohol but do this before adding any component that you think will be impacted by the washing or drying process. The rinse is important to ensure that any corrosive detergent residue is removed. If you have access to an ultrasonic cleaner even better, but if that's all too hard – if you used a 'no clean' flux, cleaning is only cosmetic.

You may wish to install the 15mm standoffs onto your main PCB's before you start to solder components onto the PCB. This provides a nice stable platform for soldering once you have turned the board over. This same approach can be used in reverse when you have finished the build and want to use the UI to test the PicoRx before you mount the project in the housing.

In this manual each time a component is identified and soldered to the PCB, the task is listed separately proceeded with a square check box or listed in a table of similar components. You can 'tick' off the components as you progress, and this keeps a record of your progress through the build. You can also check off each page as you complete the activity listed on that page:

- ☒ Component 1
- ☒ Component 2 etc.

Page | 15 ☒



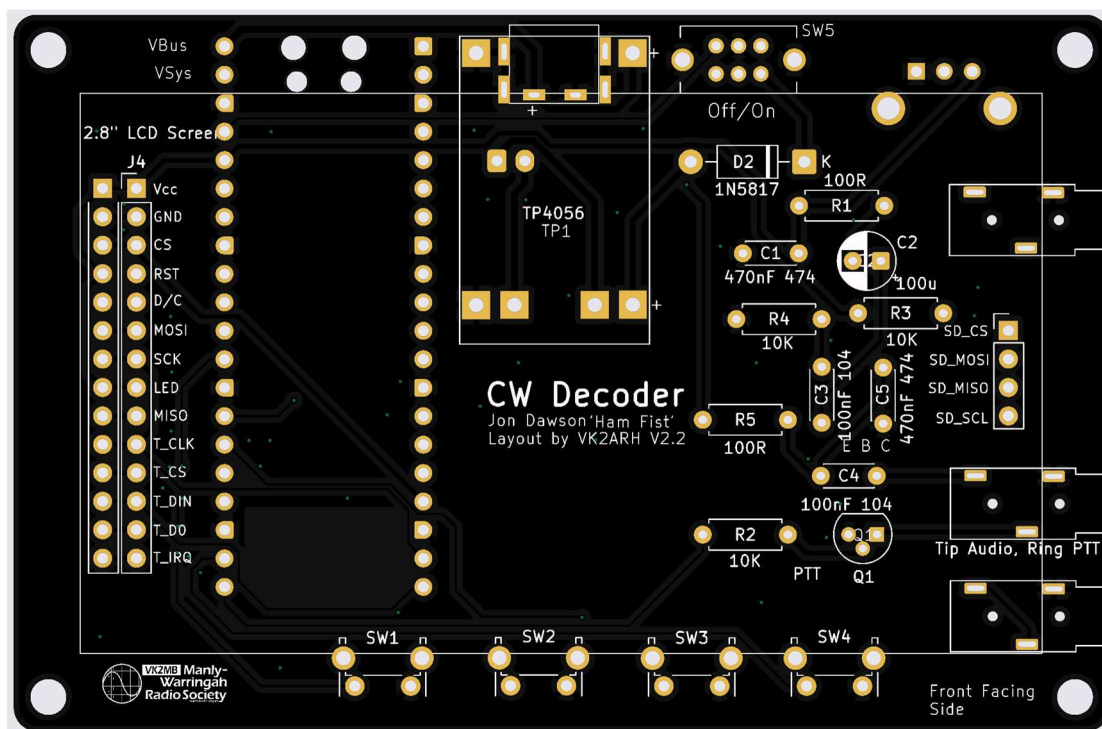
Bill of Materials (BOM)

Reference	Value	Qty
THD Resistors and Diodes		
R1,R5	100R (Brown,Black, Black,Black,Brown)	2
R2,R3,R4	10K (Brown,Black, Black,Red,Brown)	3
D2	1N5817	1
Capacitors		
C1,C5	0.47u (470n) (474)	2
C2	100u 10v	1
C3,C4	100nF (104)	2
Transistor		
Q1	BC337 or Similar	1
Switches		
SW1,SW2,SW3,SW4	Monetary Contact Micro Switch 90°	4
SW5	SW_Slide_DPDT	1
Sockets, Connectors & Volume Control		
J1, J2, J3	3.5mm Stereo Socket	3
J5	USB_C_Receptacle_PowerOnly_2P	1
J6	JST 2.0mm 2 Pin Socket	1
40 Pin Female Header Strip	Female Pin Headers 2 x 20 (Pico), 1 x 14 (J4), 1 x 4 (display) to be cut from larger 40 pin headers	2
40 Pin Male Header Strip	1 x 4, 2x2, 2x1 To be snapped off to suit mounting requirements. (May need 2 x 20 pin male headers if Pico not supplied with pins)	1.25
RV1	10K Horizontal Mounting	1
Display and Processor		
Disp_1	2.8_ILI9341_SPI_SD with Touch	1
A1	RaspberryPi_Pico V1 or V2	1
Hardware		
M3 Screws	6 or 8 mm - for top and bottom panels	8
Standoff	M3 17mm M-F	4
Standoff	M3 15mm F-F	4
PCB	Top Cover	1
PCB	Bottom Cover	1
PCB	Main PCB	1
Optional (Not included in the Kit)		
TP1	TP4056 (LiPo Managenet Module) Optional Install after J5 and J6	1
LiPo	3.7v LiPo Battery (Not Supplied)	1

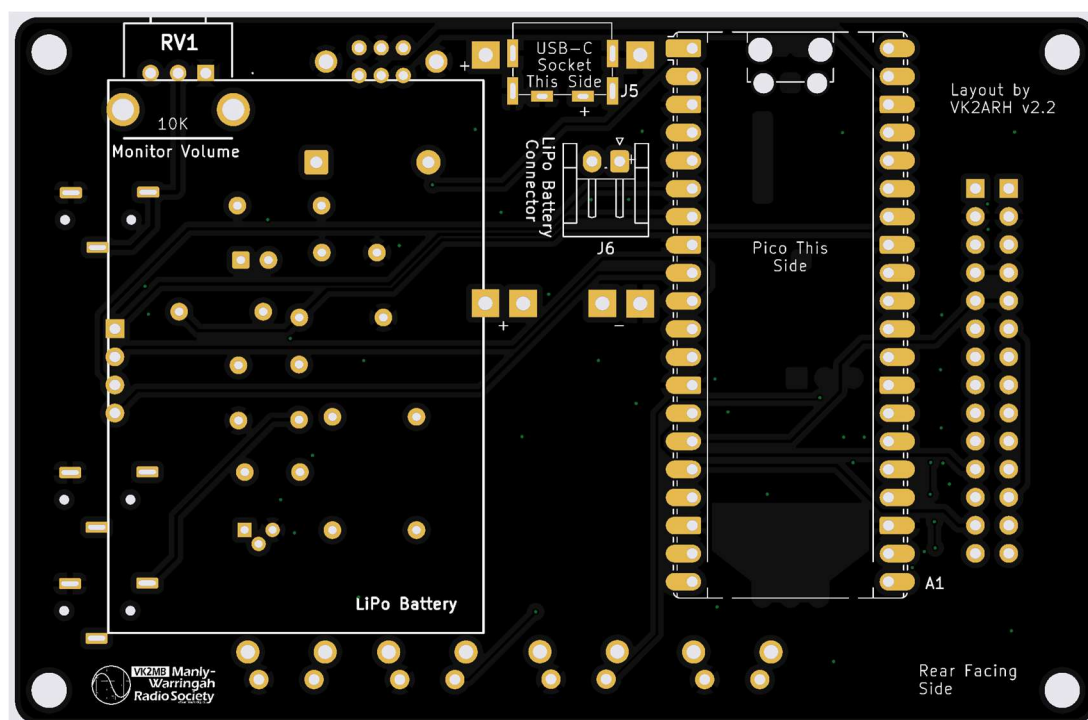


PCB and Schematic Diagram

Here is the layout of the component and rear side of the main PicoRx PCB.



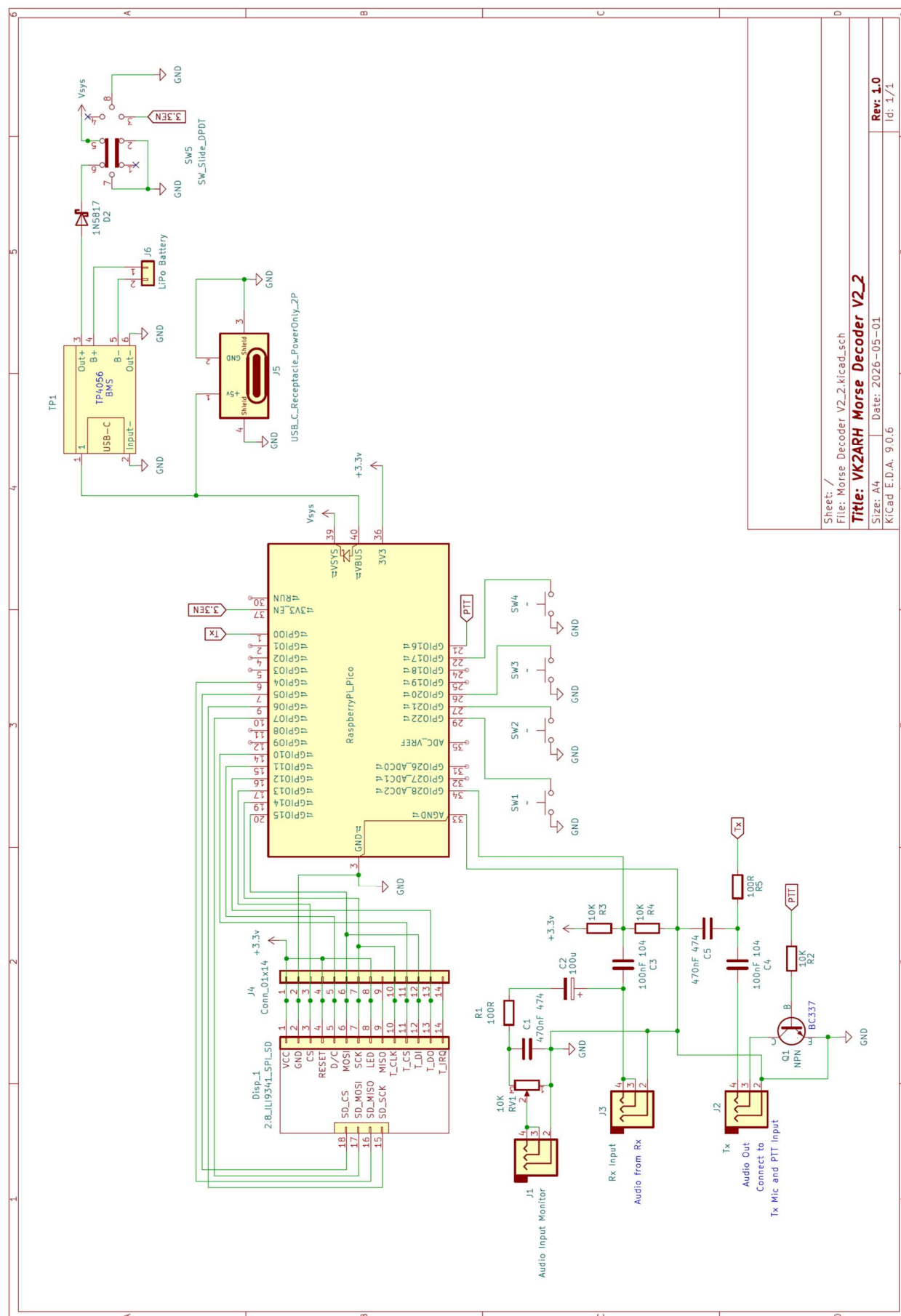
Most components are placed on the front component side of the board (above) except for the Pi Pico, USB power connectors, LiPo JST connector and the volume control. The screen print identifies which side of the board the components are mounted.



NOTE: The latest PCB is layout v2.2a and the Gerber file for the v2.2a PCB can be found on my GitHub site – there is no difference between the v2.2 board shown in this manual and the v2.2a PCB other than a few minor changes to the screen print on the main PCB. All component placement and functionality remain the same.



Morse Decoder Assembly Manual v1.0



Building the Morse Decoder

Building the decoder is relatively straight forward and the easiest way to assemble the components is to start with the smallest components first and progressively work your way through to the larger components. This approach makes handling and soldering components to the board easier. I have also arranged the components on the BOM from top to bottom in order of installation. All components are installed on the front facing side of the PCB until you get to Step 4 when some components are also installed on the rear facing side of the PCB.

Step 1: Resistors, Diode and Capacitors

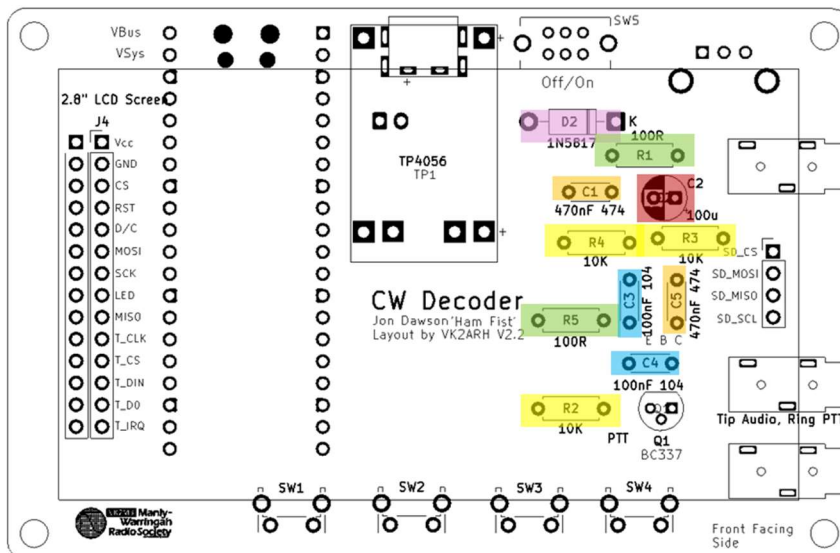
- Start with the resistors, diodes and capacitors – these are the smallest components to install. All resistors are 1/4w and should be mounted horizontally, flush with the PCB. Match the colour of the component with its position on the front side of the board, this should also align with the screen print on the board.

THD Resistors and Diodes

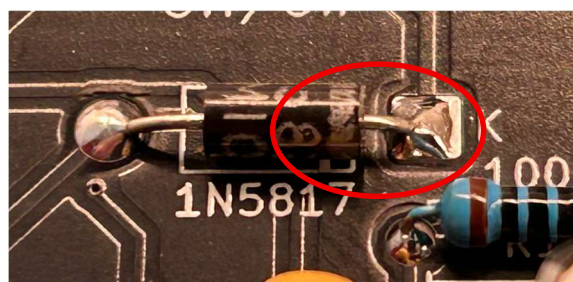
R1,R5	100R (Brown,Black, Black,Black,Brown)	2
R2,R3,R4	10K (Brown,Black, Black,Red,Brown)	3
D2	1N5817	1

Capacitors

C1,C5	0.47u (470n) (474)	2
C2	100u 10v	1
C3,C4	100nF (104)	2



- Ensure that you install the diode and electrolytic capacitor with the correct polarity. The photos below show the correct polar alignment of the electrolytic capacitor and the diode. The negative markers on the electrolytic capacitor are aligned with the shaded semi-circle on the screen print, and the band on the diode aligns with the band on the screen print with the cathode pad also indicated by a letter K on the screen print.



The -ve capacitor lead is closest to the center of the PCB whilst the band on the diode is closest to the right hand side of the PCB.

Step 2: Transistor

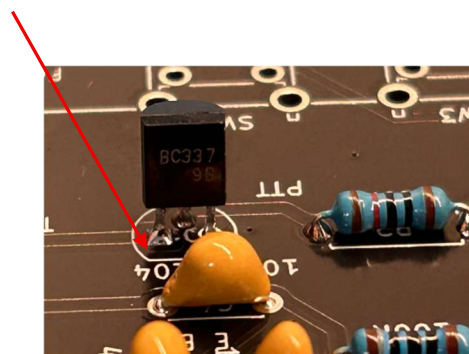
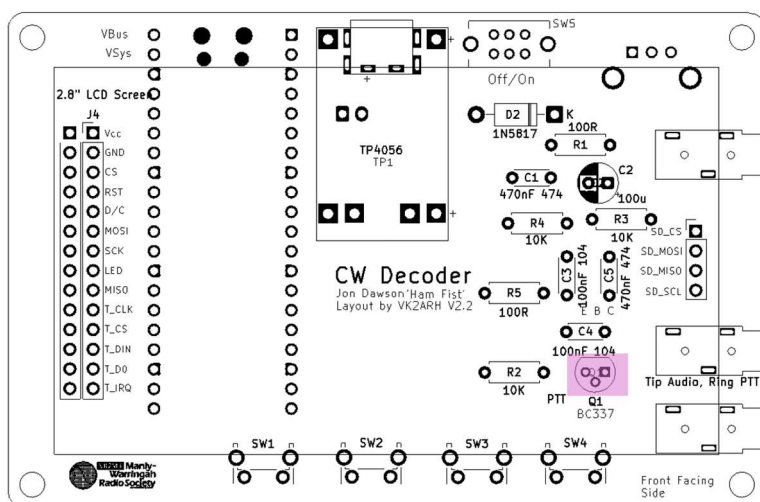
- ☐ Install the BC337 (or equivalent NPN transistor)

There is only a single transistor used in this project, and this is used for switching the PTT to ground during any transmission output. Be careful when soldering in the transistor that you do not damage the component by applying too much heat, act quickly and don't hold the soldering iron on each lead any longer than necessary. You may choose to allow each lead to cool down before moving onto another lead.

The flat face of the transistor is aligned with the flat side of the screenprint image and toward the top of the PCB.

Transistor

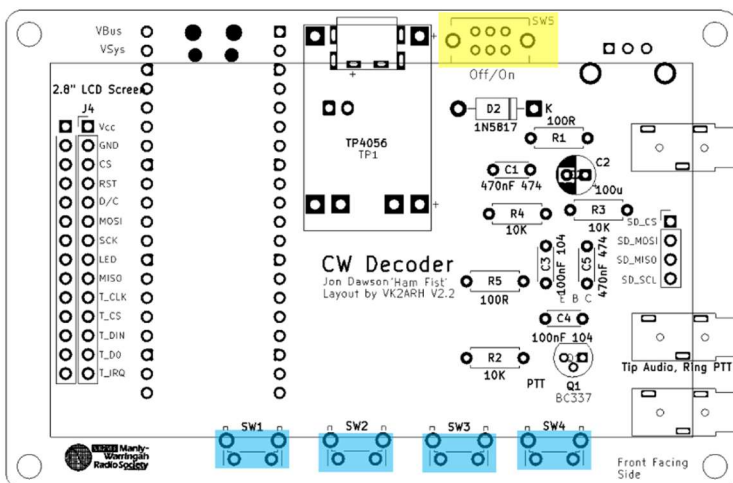
Q1	BC337 or Similar	1
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Step 3: Installing the switches

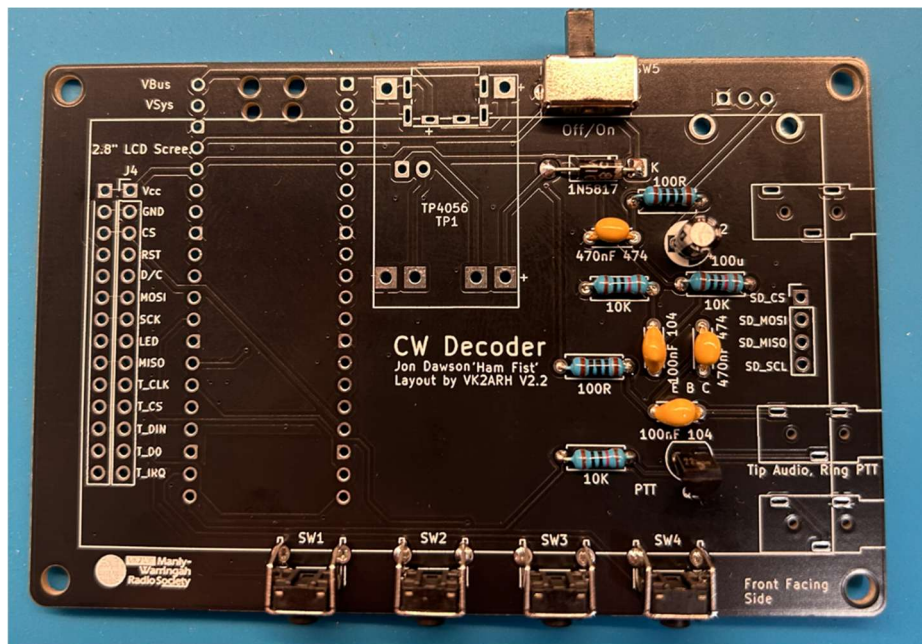
Switches

SW1,SW2,SW3,SW4	Monetary Contact Micro Switch 90°	4
SW5	SW_Slide_DPDT	1



- ☐ Install the momentary contact push button switches and the DPDT power switch in the positions shown on the diagram and identified on the PCB screen print.

Once completed the front of the main PCB should now look similar to that shown below:

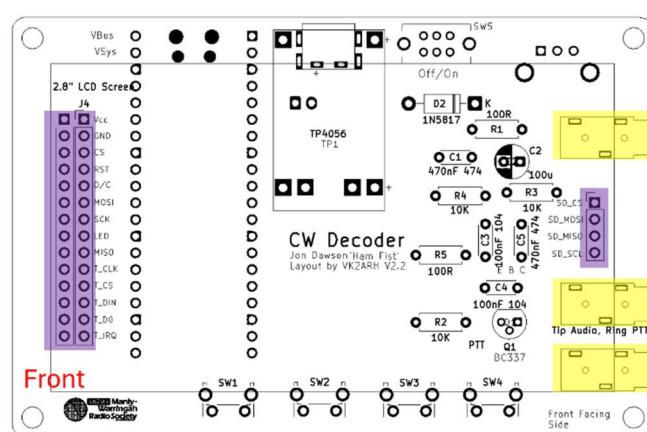
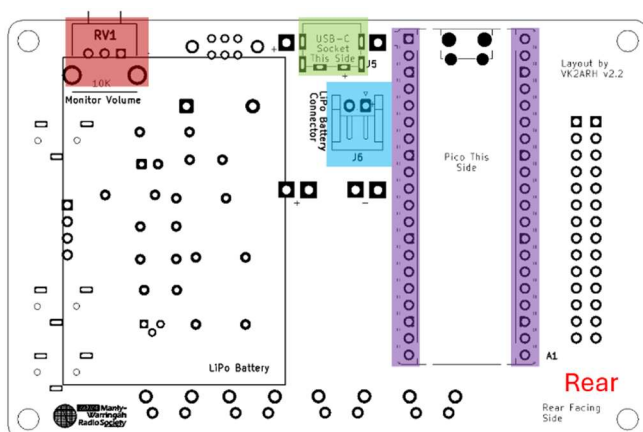


Step 4: Installing Sockets, Connectors and Volume Control

In this step components are installed on both sides of the PCB as shown in the diagram below.

Sockets, Connectors & Volume Control

J1, J2, J3	3.5mm Stereo Socket	3
J5	USB_C_Receptacle_PowerOnly_2P	1
J6	JST 2.0mm 2 Pin Socket	1
40 Pin Female Header Strip	Female Pin Headers 2 x 20 (Pico), 1 x 14 (J4), 1 x 4 (display) to be cut from larger 40 pin headers	2
40 Pin Male Header Strip	1 x 4, 2x2, 2x1 To be snapped off to suit mounting requirements. (May need 2 x 20 pin male headers if Pico not supplied with pins)	1.25
RV1	10K Horizontal Mounting	1

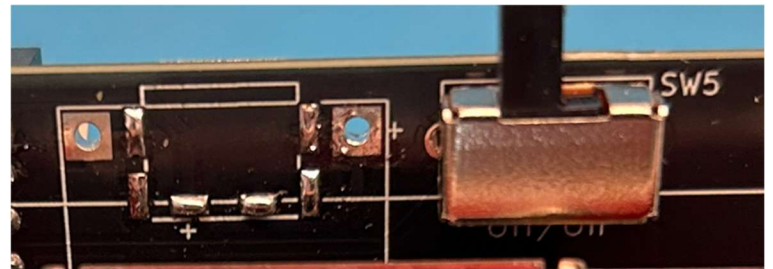
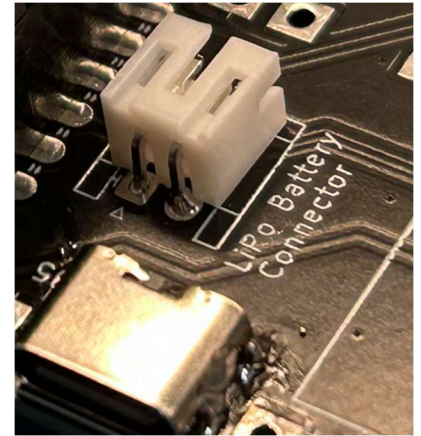
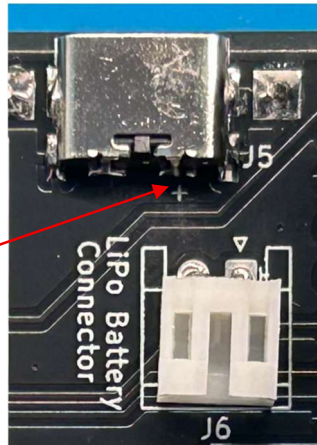


- ☐ Start by installing the 3.5mm stereo sockets (J1, J2, J3) on the front side of the PCB.
- ☐ Now turn the board over to the rear facing side and solder the USB-C power connector (J5) and the JST LiPo battery connector (J6). You only need to install the JST connector if you plan at some future stage to use it to connect a battery to the decoder. If you are installing the TP4056 LiPo Battery Management board – you may choose to not install J5 and power the decoder from the USB-C port on the TP4056. If you install both, both USB-C connectors will be wired in parallel, and this will enable you to 'share' the 5v USB-C supply with another device if you choose to do so.

When soldering the USB-C power socket J5 the leads on this socket only just protrude through to the surface of the PCB. When soldering use a fine point soldering iron tip and heat the power socket lead and the solder pad, then allow molten solder to flow into the hole and secure the socket to the board.

Check the electrical connection to the +ve power lead with a multimeter, and if necessary, solder a small 'dob' of solder to the lead on the component side to ensure adequate connection. Be sure not to apply too much heat as to melt the plastic inner of the USB-c connector.

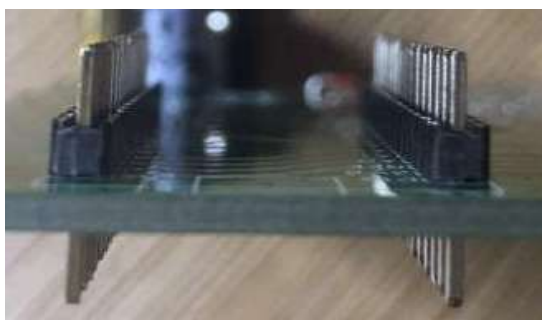
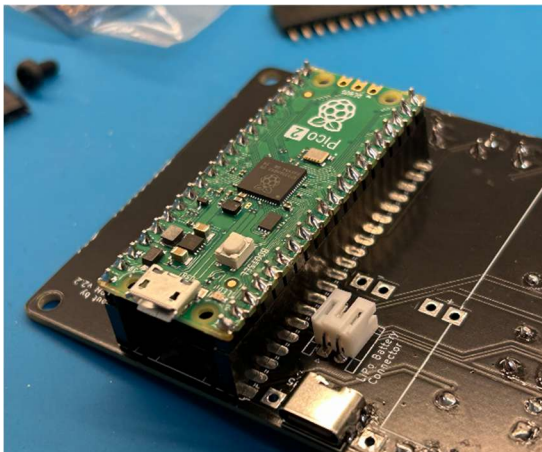
A good solder joint that has flowed down into the mounting holes will provide an electrical connection together with a strong physical attachment of the USB-C power connector to the PCB.



Prepare and install the pin headers and sockets for the Pico and the TFT display

The Pi Pico is supplied without any header pins attached so we need to attach male header pins to the Pico and female header pins to the main PCB. (Although your kit may have a Pico with the male pin headers already attached). The Pico is mounted on the underside of the board as shown in the photos below:

NOTE: Although some of these photos shows the use of a Pico 2, the decoder works just as well with a Pico 1 module.

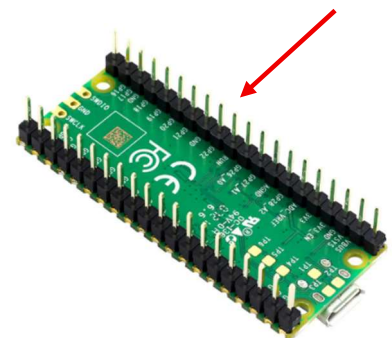


Installing the Male Pin Headers onto the Pico

□ Carefully cut a single 40 pin male header strip in half with your wire cutter to form two 20 pin headers and then solder each 20-pin header onto the Pico with the orientation shown here:

Ensure that the pins are soldered perpendicular to the Pico board.

The best way to do this is to place the pin headers into the main PCB (long lead into the board) to ensure correct alignment as shown on the left. Then sit the Pico onto the short pin headers component side up and carefully solder the **Pico to the pin headers**.



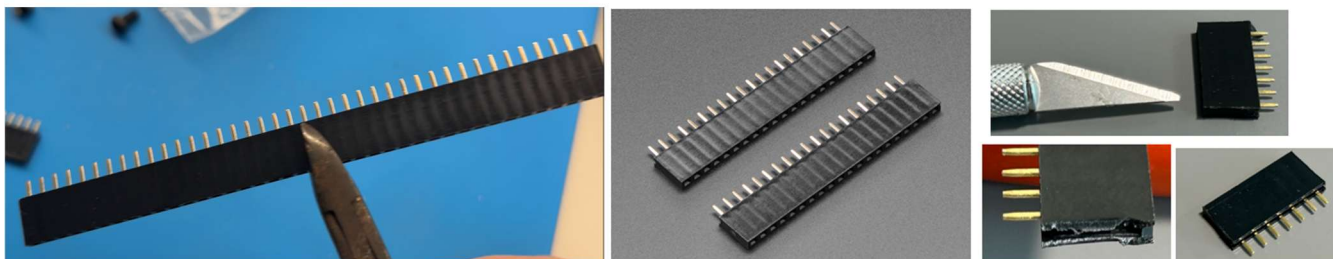
Solder quickly and spread the heat load, soldering every second pin and then returning to pick up the unsoldered pins. The pico can now be lifted from the PCB and once turned over will look like the photo above, with the long header pins facing away from the underside of the Pico board.

I strongly recommend that you **DO NOT solder the Pico pin headers directly onto the main PCB** but install female pin headers on the main PCB to house the Pico, this gives you the flexibility to easily replace the Pico should it be required.

Preparing the Female Pin Header Strips

Header strips can be cut down from a longer pin header to form a shorter lower pin count header. This technique is used to prepare the 20 pin headers used for the Pico and the remaining 19 pin header will be used to manufacture the 14 pin and 4 pin headers used for mounting the TFT display. Manufacture the two 20 pin headers for the Pico first.

Align the strip up against the 20 holes for the Pico on the main PCB and then score the pin header where it is to be cut. Double count the number of pins to be left on the header to ensure that you have 20 pins before you make the cut. Take your wire cutters and cut the strip at the score mark. Hint: if you cut the 20 pin strip from one end of the original 40 pin header, you will have one nice clean end on the pin header which you can solder toward the edge of the PicoRx and this will look a lot neater.

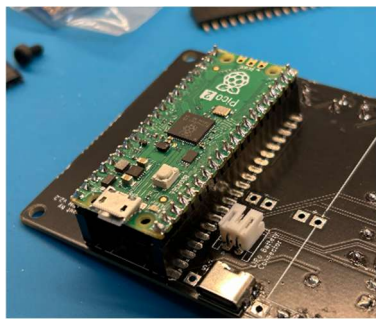
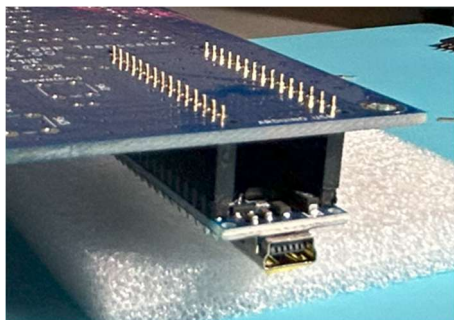


When finished your strip will end up looking like those shown above center. The cutting process will destroy one of the pin headers and leave a ragged edge, but this is cleaned up easily with a modeling knife and/or sandpaper as shown by example in the right hand photos above.

If using a standard 40 pin header strip, you will end up with a 20 pin and a 19 pin female header. As the Pico requires a second 20 pin header, you will need to use the second 40 pin strip and cut the 20-pin section from it. After sacrificing the 21st pin on each strip, you will be left with two 19 pin strips. One will be used to manufacture the 14 pin and 4 pin female headers used to mount the TFT Display after sacrificing the 15th pin in the header. You will then be left with a spare 19 pin female header which can be repurposed on another project.

Installing the Pico Pin Header

- ☐ Install your pin headers onto the rear side of the PicoRx PCB. Fit the pin headers and the Pico onto the board, turn the board upside down and place a piece of foam or similar underneath the Pico so that the weight of the board allows the pins to protrude through the solder side of the board. This ensures vertical alignment of the headers. Now solder the pin headers to the PCB. Act quickly as the heat from the soldering iron can easily melt the header. Consider spacing out the soldering by missing every second pin and then returning to solder the remaining pins and in doing so spread the thermal impact on the header pins. The finished results should look similar this even though the example photo on the left shows is an Arduino Nano – the same principle applies to the Pico. The photo on the right shows the Pico installed onto the decoder board.

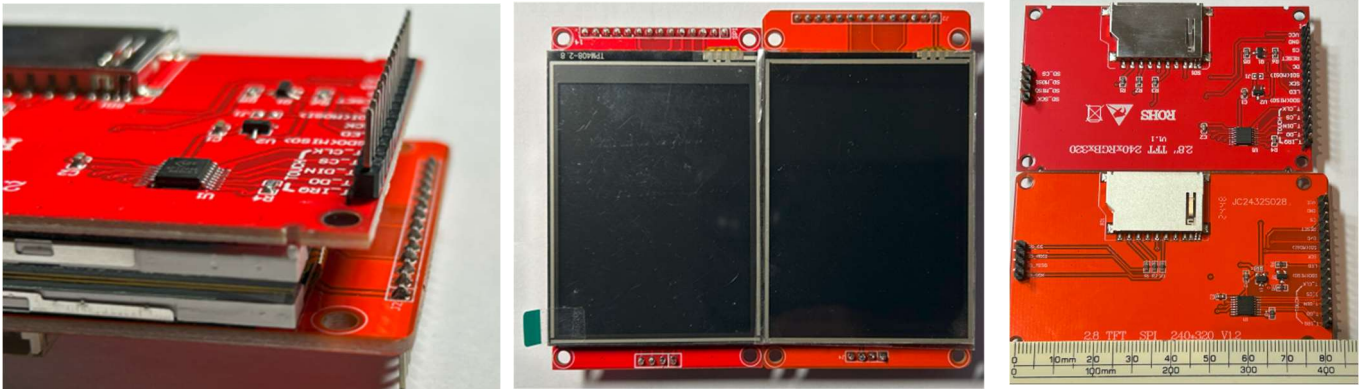


The photo on the left illustrates the process using an Arduino Nano, but the same principle applies to the Pi-Pico shown on the right.

I recommend removing the Pico at this stage and leaving it off the PCB until you have finished your build and undertaken 'power on' test and configured the software before inserting the Pico back onto the main PCB.

Installing the Pin Headers for the TFT Display

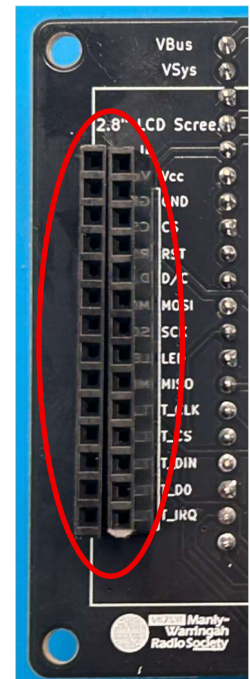
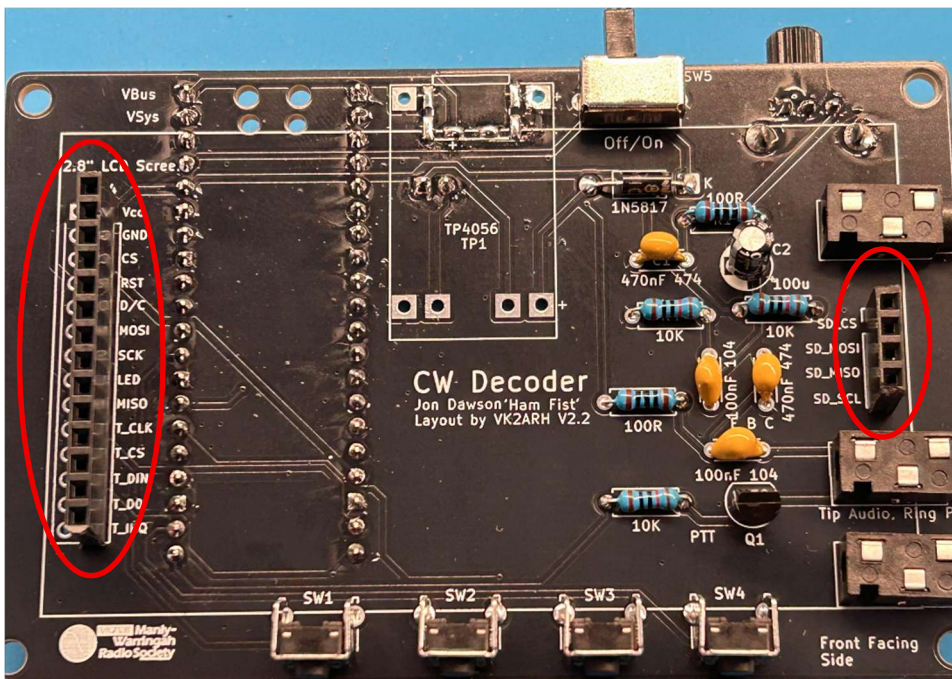
Different manufacturers provide 2.8" TFT screens with different 14 pin connector positions, the version 2.2 board provides two sets of 14 pin connectors to enable use of either of the common pin configurations found on the 2.8" TFT displays to be used



Different TFT Displays have the 14 pin header in different locations. The longer board is the original factory specification.

You will need to cut the 14 pin and 4 pin female headers from the 19 pin female header left over when cutting the 20 pin female pin headers to mount the Pico. The easiest way to do this is to sacrifice the 15th pin (and 5th pin for the 4 pin header) when cutting off the header and then file down or trim the end of the header to neaten it up.

- Install the 14 pin female header in the connector which aligns with your TFT display.
- Install the 4 pin female header on the opposite side of the board.

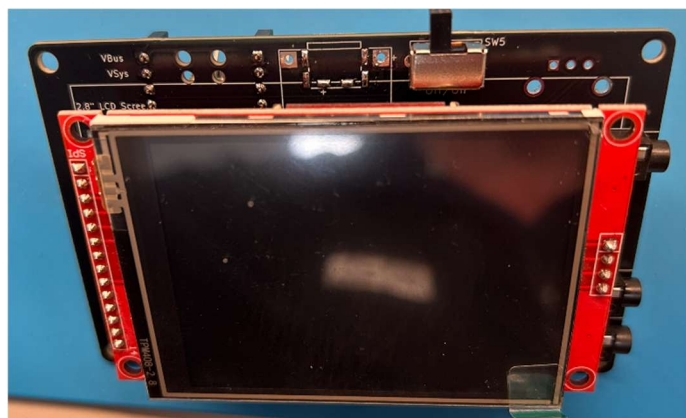


After installation of the TFP support headers the front of the PCB should look similar to that shown above left. This example shows installation of the 14 pin header to support the 'shorter' 2.8" display. You may wish to install a second 14 pin header using the spare 19 pin header to manufacture it, so that you have the option going forward to support either the longer or shorter board as shown in the above right photo.

Ensure that the pin headers are perpendicular to the PCB – the easiest way to align these is to solder a single pin to the board – check alignment and if adjustment is needed, reheat and melt the soldered pin, adjust the position of the header and once in position proceed to solder the remaining pins to the board.

- ☐ Install the male 4 pin header onto the TFT display to complete the TFT's mounting requirements. Do this by carefully cutting off a 4 pin strip from the larger male pin header strip.

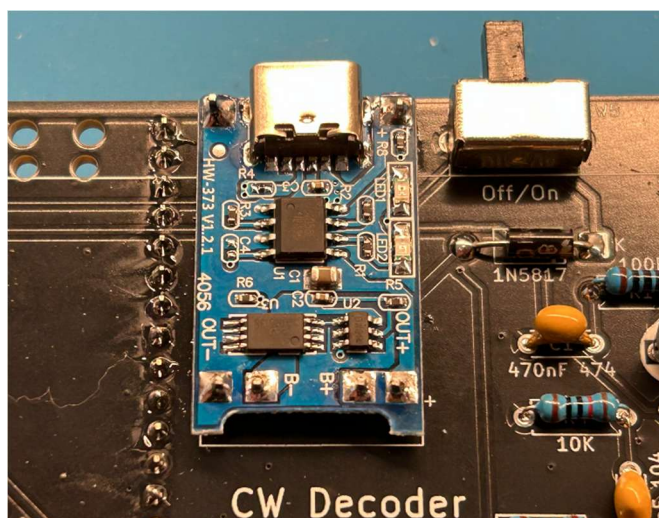
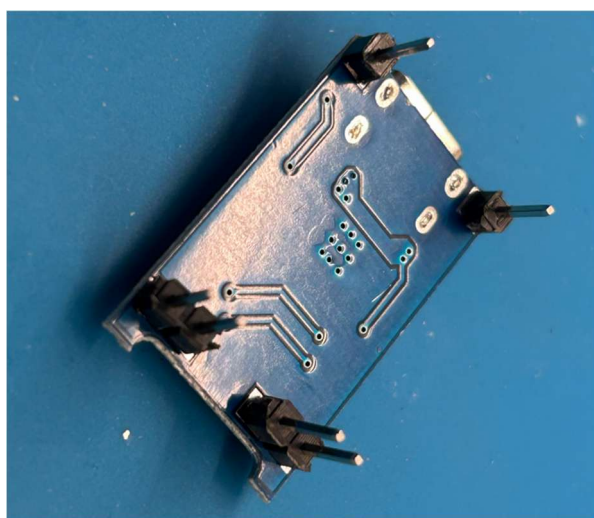
The finished TFT display installation will look similar to this:



Step 5: Installing the Optional TP4056 LiPo Battery Module

Your kit may have been supplied with the optional TP4056 Lithium Battery Management Module. Even if you don't wish to immediately install a LiPo battery at this stage, now is a good time to install the module to make provision for future use. This module is soldered directly to the PCB using male header pins.

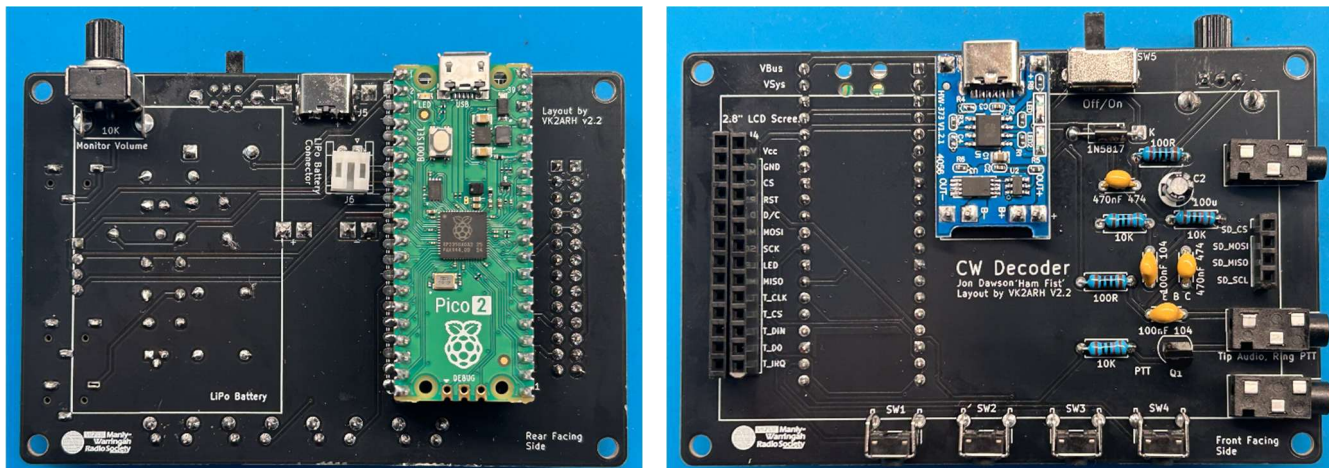
Whilst the board has two sets of 2 pins on the output end of the board, these are approx. 3mm apart so a pair of male 2.54mm pin headers is not quite wide enough to fit into the PCB mounting holes, but it can be 'forced'. The module is best installed using single male pin headers broken off from the large male pin header strip. Start by placing single male pin headers into the front side of the PCB (long side into the PCB) and then insert the TP4056 module onto the headers and solder them to the module. Turn the board upside down and solder the pin headers to the board from the rear side.



NOTE: If you are installing the TP4056 it is not necessary to install the USB-C power socket

This completes the construction of the decoder PCB. ☺

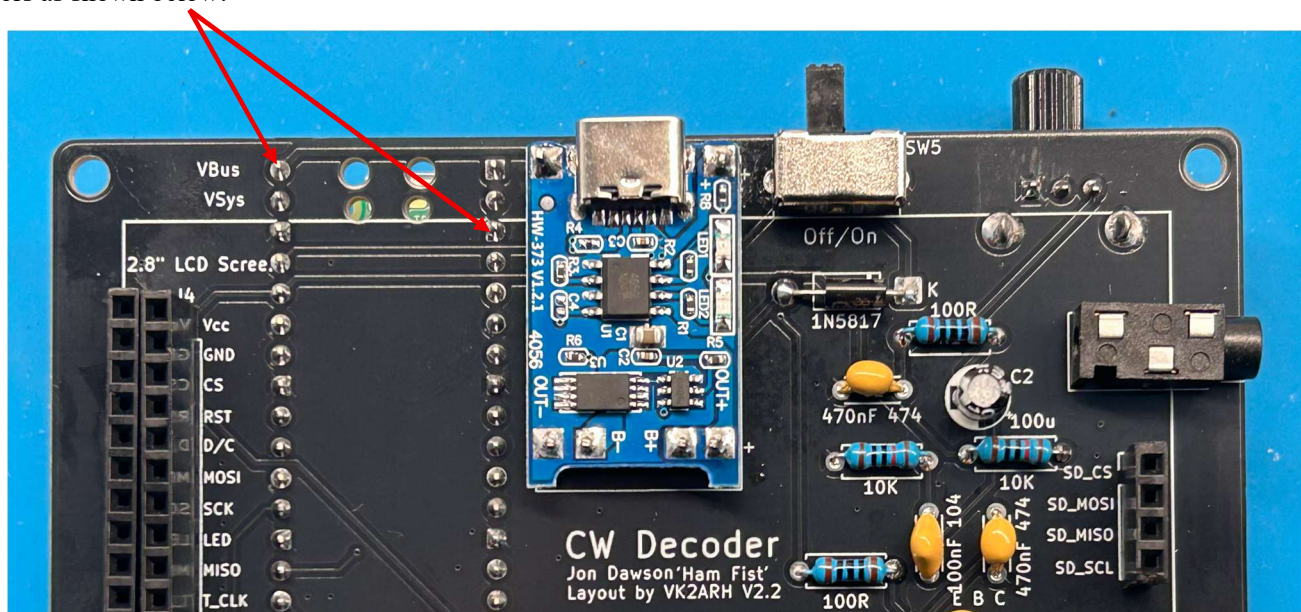
The finished board should now look similar to those shown below:



Step 6: Pre Start Up Power Testing

- ☐ **Before powering up the PCB remove the Pico processor from the board** (if you haven't already done so) and perform a continuity check on the DC inputs to ensure that you have no shorts.

This can be achieved by measuring the resistance between VBus and Ground connection points on the external connectors as shown below:

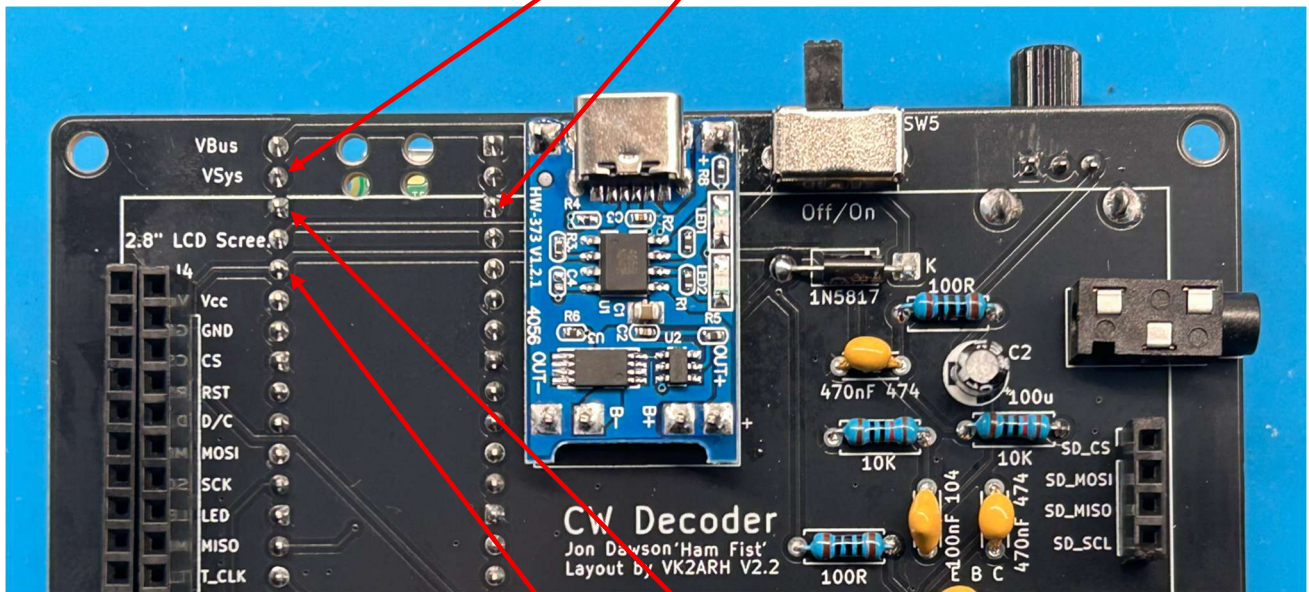


- ☐ Now connect a USB-C power supply to the USB-C socket and validate that you have 5v between the same two connection points.

As an extra precaution during the first power up, if you have access to a current limiting power supply, connect the PCB to a 5v source supplying the USB-C plug after setting the current limit to 50mA. The PicoRx will draw less than 40mA without the TFT Screen

When you add the optional external 2.8" Colour TFT display the Decoder will draw approx. 90mA. You will need to adjust the power supply current limit to approx. 120mA when you are ready to test the installation of the TFT display.

- ☐ Insert your Pico processor and connect power again to the USB-C connector, switch the power switch to ON and check that you have approx. 4.2v across the Vsys and GND measurement points you used earlier to check for shorts.



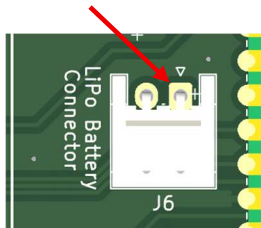
- ☐ Check that you have 3.3v being supplied to the VCC pin and Gnd on the TFT Display
- ☐ Turn the power switch OFF , remove the USB-C power supply and proceed to install the software onto your Pico processor.

NOTE: When power is connected to the PCB which has the TP4056 module installed without a battery connected, the TP4056 module's blue light will illuminate, and the red LED will flash. When a 3.7v LiPo battery is installed, the red LED will illuminate indicating that the battery is charging. When the battery is fully charged the red light will extinguish and the Blue LED will illuminate.

Step 7: Installing a LiPo Battery (Optional)

Adding a LiPo Battery provides a good deal of flexibility in the use of the decoder although not necessary if you always have access to a 5v USB cable. Before connecting a LiPo battery to the decoder you should undertake the following:

- ☐ Ensure that you have trimmed down all soldered leads that may be sticking out from the PCB where the battery is to be installed to ensure that they do not puncture the battery. I attach the battery to the decoder using double sided tape and it can be mounted either directly onto the PCB or onto the rear panel PCB when housing in the 'PCB Sandwich' housing.
- ☐ Ensure that you have the correct polarity in the battery JST connector before plugging into the board. The +ve connection is shown with the inverted triangle and square solder pad on the PCB.



The battery plug connection is shown on the right. It is possible to switch the battery wires by carefully lifting the plastic locking lug on the plug and removing the wires, swapping them over and inserting them into the opposite position in the plug. Take care not to short out the battery should you need to do this:



Installing the Software

Loading the firmware into the Pico is achieved as follows:

- ☐ Download Jon's firmware from: <https://github.com/dawsonjon/HamFist>

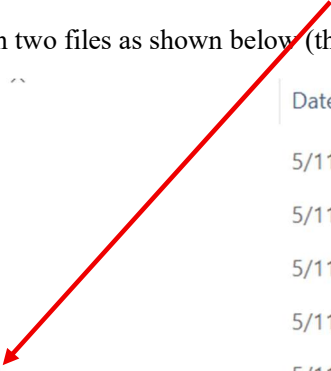
Unzip the package and under the build folder you will find directories for each of the Pico types (original, Pico2, Pico2w and PicoW).

Name	Date modified	Type
rp2040.rp2040.rpipico	5/11/2026 9:41 PM	File folder
rp2040.rp2040.rpipico2	5/11/2026 9:41 PM	File folder
rp2040.rp2040.rpipico2w	5/11/2026 9:41 PM	File folder
rp2040.rp2040.rpicow	5/11/2026 9:41 PM	File folder

Choose the appropriate binary for either the original Pico or the Pico2 depending upon which Pico you are using. Both work fine but need to be installed on the correct Pico variant. Within the chosen directory you will find a number of files, but the one you are looking for is the use the `cw_decoder_xxx.uf2` (where xxx corresponds to the type of Pico board you are using).

- ☐ which should contain two files as shown below (the example shows Jon Dawson's Pico2 version)

Name	Date modified	Type	Size
 cw_decoder.ino.bin	5/11/2026 9:41 PM	BIN File	443 KB
 cw_decoder.ino.elf	5/11/2026 9:41 PM	ELF File	2,888 KB
 cw_decoder.ino.map	5/11/2026 9:41 PM	MAP File	1,422 KB
 cw_decoder.ino.uf2	5/11/2026 9:41 PM	UF2 File	887 KB
 cw_decoder_pico2.uf2	5/11/2026 9:41 PM	UF2 File	886 KB



- ☐ Connect a USB data cable from your computer to the Pico while holding in the pushbutton by the USB connector on the Pico and then release the button. The Pico should show up as a new USB drive on your computer .
- ☐ Copy the required uf2 file you downloaded earlier onto the USB drive, the Pico will restart and you should be ready to start the decoder.

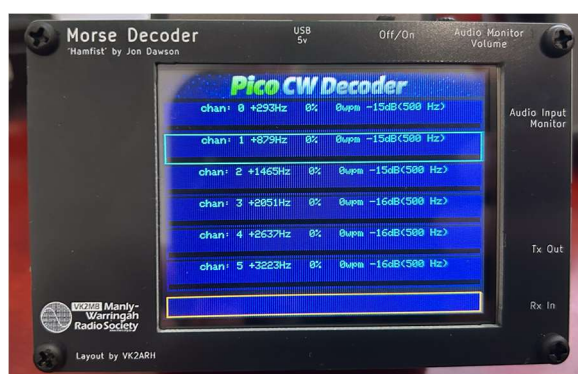
NOTE: At the time of writing Joh's new code also supports using the touch screen capability, however this was not coded in the builds made available on his site. If you wish to use the touch screen capability you will need to compile the source code with the appropriate definition for the touch screen and upload it to your Pico. Details of this process can be found on Jon's GitHub site.

Jon's original source is available from the same link shown above for those who wish to modify enhance and develop their own software further.

Checking Decoder Operation

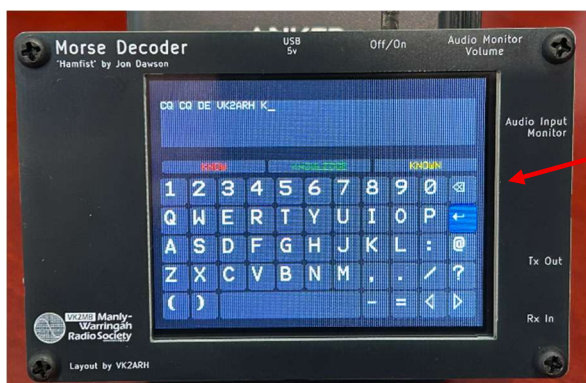
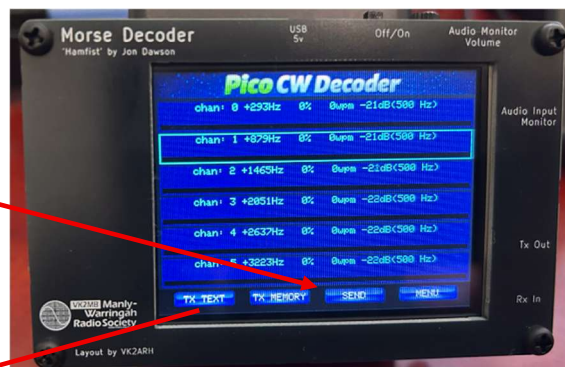
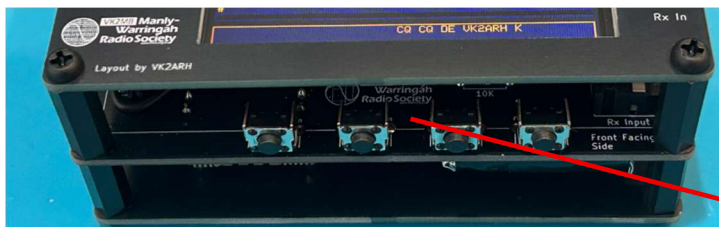
- ☐ Once the software is loaded, power supply testing completed and power turned off, install the Pico onto the main board, along with the TFT display.
- ☐ You may find handling and testing the Decoder easier if you temporarily install the 4 x M3 x 15mm F-F spacers on the back of the PCB using the screws provided and then turn the PCB face up. This will give you a stable platform to configure and test your project before installing it into the housing.

- ☐ Power on the Decoder using the USB-C 5v power supply. After a short pause – you will see the splash screen shown below left which after a short pause will move to the standard front screen below:



If you have enabled use of the touch functionality – you will first see the touch calibration screen which requires you to calibrate the screen by pressing on the small circle as it moves around the screen before moving to the standard front screen above.

Experiment with each of the push button switches on the lower side of the Decoder. Click any switch to activate the button menu and then click the switch immediately below that menu item to activate that function. If you know which function you are wanting to enter, simply double click the required button.



If you have activated the touch screen in the code – you will be able to bring up the data entry screen shown below by double clicking the TX_TEXT button (the fastest way to access it) – check that the touch screen is working correctly.

If successful you will have confirmed that the Decoder has the software successfully loaded and is working correctly. If you do not see the correct operation of the Decoder, check that you have successfully downloaded the software and go back to basics and start 'fault finding' your build.

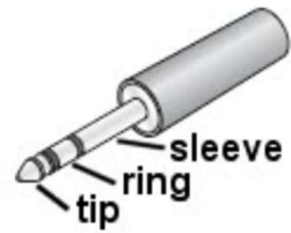
Check the correct installation of components and that they are soldered without solder bridges, dry joints or shorts. The board has been built and tested several times, so any problems with the operation of the board at this stage are likely unique to the individual build (or the components you used) and beyond the scope of this manual to diagnose. Good luck with your diagnostic journey.

Connecting the Audio Cables

Audio to and from the decoder is via a standard 3.5mm audio cable. For the Rx Audio In and Audio Input Monitor sockets the tip and ring are joined – effectively providing a mono input and output, the sleeve forms the ground connection.

For the Tx Audio Out the ring is used for PTT. When the decoder is being used to transmit CW, the Audio out is presented to the tip connection and the ring is grounded – providing an ability to key a remote transmitter if required.

When you plug an audio cable into your receiver you generally lose the ability to hear the received signal, so the Audio Input Monitor socket provides a parallel output to the Rx Audio In which can be fed to an external speaker or headphone to monitor the incoming audio. The volume control on the decoder varies the volume of this output.



Housing the Morse Decoder

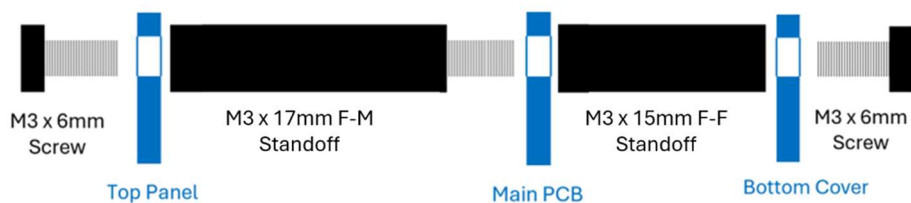
Assembling the Morse Decoder into the 'PC Sandwich' housing result in practical and attractive housing for the project. The finished project looks something like that shown below.



The housing is put together using the following:

- Top Cover
- Main PCB
- Bottom Cover
- 8 x M3 6mm or 8mm Nylon (or Stainless Steel) Machine Screws
- 4 x M3 17mm M-F Nylon Insulated Standoff's
- 4 x M3 15mm F-F Nylon Insulated Standoff's

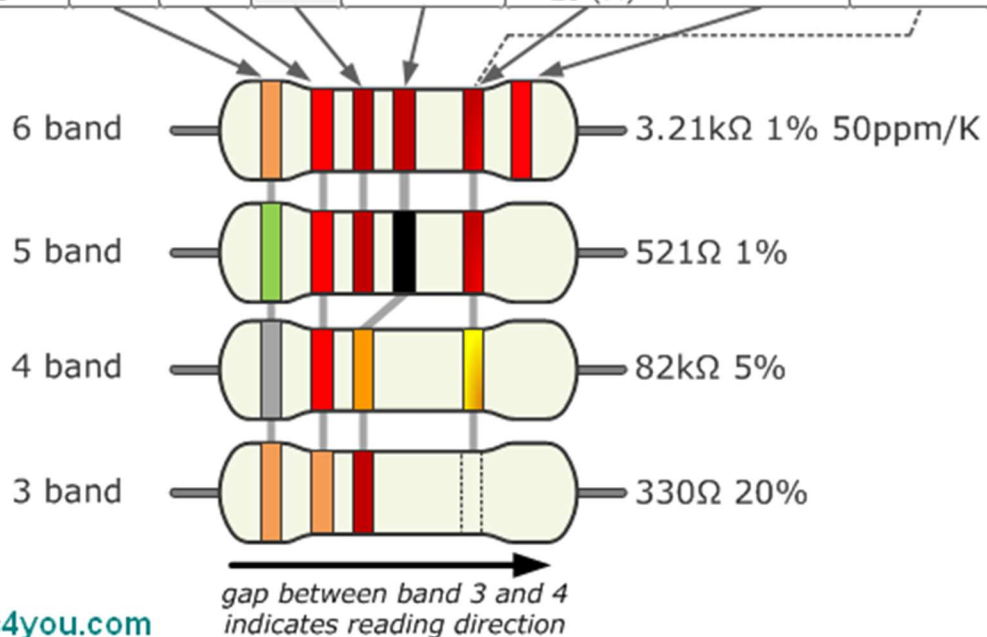
The housing is assembled on each corner using the component side view as shown below. Start by installing the Main PCB between the two standoffs, then add the Top Panel and Bottom Cover securing them with the screws.



Resistor, Capacitor and Inductor Identification Charts (For Info)

Resistor Color Code Chart

Color	Significant figures			Multiply	Tolerance (%)	Temp. Coeff. (ppm/K)	Fail Rate (%)
black	0	0	0	x 1		250 (U)	
brown	1	1	1	x 10	1 (F)	100 (S)	1
red	2	2	2	x 100	2 (G)	50 (R)	0.1
orange	3	3	3	x 1K		15 (P)	0.01
yellow	4	4	4	x 10K		25 (Q)	0.001
green	5	5	5	x 100K	0.5 (D)	20 (Z)	
blue	6	6	6	x 1M	0.25 (C)	10 (Z)	
violet	7	7	7	x 10M	0.1 (B)	5 (M)	
grey	8	8	8	x 100M	0.05 (A)	1(K)	
white	9	9	9	x 1G			
gold			3th digit only for 5 and 6 bands	x 0.1	5 (J)		
silver				x 0.01	10 (K)		
none					20 (M)		



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Capacitor Identification Charts and Letter Code Table

Capacitors

Ceramic Capacitor

Symbol (Non-Polarized)

Electrolytic Capacitor

Symbol (Polarized)

Microfarads (μF)	Nanofarads (nF)	Picofarads (pF)
0.000001 μF	0.001 nF	1 pF
0.00001 μF	0.01 nF	10 pF
0.0001 μF	0.1 nF	100 pF
0.001 μF	1 nF	1,000 pF
0.01 μF	10 nF	10,000 pF
0.1 μF	100 nF	100,000 pF
1 μF	1,000 nF	1,000,000 pF
10 μF	10,000 nF	10,000,000 pF
100 μF	100,000 nF	100,000,000 pF

Code	Max. Voltage
1H	50V
2A	100V
2T	150V
2D	200V
2E	250V
2G	400V
2J	630V

Code	Percentage
B	± 0.1 pF
C	± 0.25 pF
D	± 0.5 pF
F	± 1%
G	± 2%
H	± 3%
J	± 5%
K	± 10%
M	± 20%
Z	+80%, -20%

Capacitor Letter Codes Table

Picofarad (pF)	Nanofarad (nF)	Microfarad (uF)	Code	Picofarad (pF)	Nanofarad (nF)	Microfarad (uF)	Code
10	0.01	0.00001	100	4700	4.7	0.0047	472
15	0.015	0.000015	150	5000	5.0	0.005	502
22	0.022	0.000022	220	5600	5.6	0.0056	562
33	0.033	0.000033	330	6800	6.8	0.0068	682
47	0.047	0.000047	470	10000	10	0.01	103
100	0.1	0.0001	101	15000	15	0.015	153
120	0.12	0.00012	121	22000	22	0.022	223
130	0.13	0.00013	131	33000	33	0.033	333
150	0.15	0.00015	151	47000	47	0.047	473
180	0.18	0.00018	181	68000	68	0.068	683
220	0.22	0.00022	221	100000	100	0.1	104
330	0.33	0.00033	331	150000	150	0.15	154
470	0.47	0.00047	471	200000	200	0.2	204
560	0.56	0.00056	561	220000	220	0.22	224
680	0.68	0.00068	681	330000	330	0.33	334
750	0.75	0.00075	751	470000	470	0.47	474
820	0.82	0.00082	821	680000	680	0.68	684
1000	1.0	0.001	102	1000000	1000	1.0	105
1500	1.5	0.0015	152	1500000	1500	1.5	155
2000	2.0	0.002	202	2000000	2000	2.0	205
2200	2.2	0.0022	222	2200000	2200	2.2	225
3300	3.3	0.0033	332	3300000	3300	3.3	335

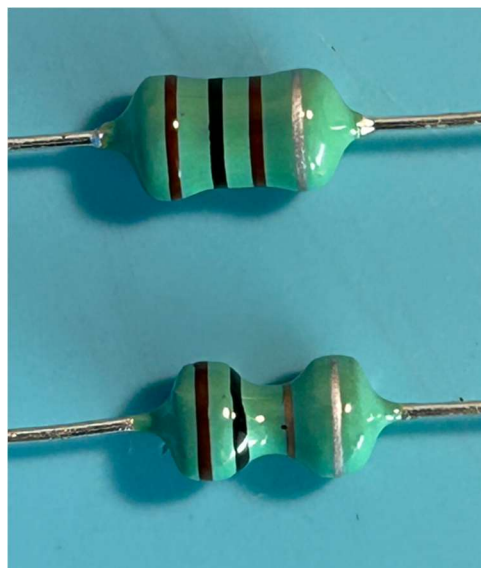
Inductor Identification Chart

4-band-code

	0	1	20%
Silver			0.01
Gold			0.1
Black			1
Brown	1	1	10
Red	2	2	100
Orange	3	3	1000
Yellow	4	4	
Green	5	5	
Blue	6	6	
Violet	7	7	
Grey	8	8	
White	9	9	

tolerance

multiplier



Brown: Black: Brown: Silver
 1: 0: x 10
 (multiplier)
 = **100uH** 10% tolerance

Brown: Black: Gold: Silver
 1: 0: x 0.1 (multiplier)
 = **1 uH** 10% tolerance

